Ising relaxation, local parent Hamiltonians and clock normalization

The periodic zero-field Ising heat-bath chain has the exact full relaxation gap $a[1-\tanh(2b)]$ for every $N \geq 3$, at per-site refresh rate $a > 0$ and finite ferromagnetic coupling $b \geq 0$. Its square-root Gibbs transform is an explicit positive three-site local Hermitian operator with that gap. Multiplication by a supplied action $K > 0$ gives an energy gap K times the rate. The invariant law fixes neither clock nor action normalization. A volume-uniform relaxation gap also coexists with a growing global mixing time. These are three different statements.

The exact Ising gap and detailed-balance parent construction are established results. The proof below is self-contained, with C124/C126 source matches in B68, B71 and B73. Historical model/coordinator checks are internal; no novelty is claimed.

Gibbs law, refresh convention and the full generator

On $\Omega_N = \{-1,1\}^N$ with cyclic indices, $N \geq 3$, put

$$\pi_{N,b}(\sigma) = Z^{-1} \exp\left(b \sum_i \sigma_i \sigma_{i+1}\right), \qquad b = \beta J, \quad \theta = \tanh(2b).$$

J has energy units, β inverse energy and b is dimensionless. Independent rate-a site clocks refresh from the conditional Gibbs law:

$$p_i(\sigma) = \frac{1}{2} + \frac{\theta}{4}(\sigma_{i-1} + \sigma_{i+1}), \qquad Q_N = a \sum_i (E_i - I).$$

A refresh may leave the spin unchanged. The actual flip rate is $a[1 - \theta\sigma_i(\sigma_{i-1} + \sigma_{i+1})/2]/2$. Conditional expectations E_i are orthogonal projections in $L^2(\pi)$, so Q_N is self-adjoint. Finite b makes every flip probability positive and the chain irreducible. The domain is the entire finite function space; its centered-space gap has inverse-time units in the specified sampling clock.

A calibrated velocity $v_i = u\sigma_i$, $u > 0$, is centered and has variance u^2. The full vector identifies the configuration, but its linear span does not cover every centered function; the following proof controls all modes.

Exact all-mode gap

Couple two chains with the same clocks and refresh uniforms. With disagreement indicators D_i and Hamming distance $D = \sum_i D_i$, the refresh mismatch probability is bounded by $\theta(D_{i-1} + D_{i+1})/2$. Hence

$$\mathcal{L}D \leq -aD + \frac{a\theta}{2} \sum_i (D_{i-1} + D_{i+1}) = -a(1-\theta)D.$$

Finite-state evolution and Gronwall give $\mathbb{E}D_t \leq e^{-rt} D_0$, $r = a(1-\theta)$. Therefore every function's Hamming Lipschitz seminorm obeys $\mathrm{Lip}(P_t f) \leq e^{-rt}\,\mathrm{Lip}(f)$. A nonconstant real eigenfunction with $-Q_N f = \lambda f$ has positive finite seminorm, so cancellation gives $e^{-\lambda t} \leq e^{-rt}$ and $\lambda \geq r$. Reversibility supplies a real eigenbasis; the estimate covers every centered eigenmode without a size prefactor.

For $M = \sum_i \sigma_i$,

$$Q_N \sigma_i = -a\sigma_i + \frac{a\theta}{2}(\sigma_{i-1} + \sigma_{i+1}), \qquad Q_N M = -a(1-\theta)M.$$

M is centered and nonzero because every configuration has positive weight. This matching eigenfunction proves

$$\boxed{\operatorname{gap}(-Q_N) = a[1 - \tanh(2b)].}$$

For bounded mean velocity $V = uM/N$, $\chi(V) = \operatorname{Var}_\pi(V)/\operatorname{gap}(-Q_N)$. Its variance is positive at every finite N without a claimed uniform lower bound; $\chi(V)/\operatorname{Var}(V)$ detects the slow-mode time. The action-valued response $2m\chi(V)$ additionally supplies a mass and velocity calibration, as in susceptibility.

The square-root Gibbs parent and its local terms

Let $D = \operatorname{diag}\pi$ and $U : L^2(\pi) \to (\mathbb{C}^2)^{\otimes N}$, $(Uf)_\sigma = \sqrt{\pi(\sigma)}f(\sigma)$. This is unitary between the weighted and counting inner products. Define

$$A_N = -UQ_NU^{-1} = -D^{1/2}Q_ND^{-1/2}.$$

Detailed balance gives Hermiticity, and the exact Dirichlet form is

$$\langle\psi, A_N\psi\rangle = \frac{1}{2}\sum_{\sigma,\eta}\pi(\sigma)Q_{\sigma\eta}\left|\frac{\psi_\eta}{\sqrt{\pi(\eta)}} - \frac{\psi_\sigma}{\sqrt{\pi(\sigma)}}\right|^2 \geq 0.$$

Irreducibility makes the kernel one-dimensional, spanned by

$$|\Omega\rangle = Z^{-1/2}\sum_\sigma e^{(b/2)\sum_i \sigma_i\sigma_{i+1}}|\sigma\rangle.$$

Let Z_i be the diagonal spin and X_i its flip. Put $s_0 = \operatorname{sech}(2b)$ and $c_\pm = (s_0 \pm 1)/2$. Then

$$A_N = \sum_i h_i, \qquad h_i = \frac{a}{2}\left[I - \frac{\theta}{2}Z_i(Z_{i-1} + Z_{i+1}) - (c_+I + c_-Z_{i-1}Z_{i+1})X_i\right].$$

For fixed neighbour sum $S = 0, \pm 2$, the transformed off-diagonal flip entry is $-\sqrt{r_i(\sigma)r_i(\sigma^i)} = -a/[2\cosh(bS)]$; its diagonal is unchanged. The coefficient is exactly $c_+ + c_-\sigma_{i-1}\sigma_{i+1}$, proving the formula. The neighbours are distinct for $N = 3$ as required. Alternatively $h_i = a(I - UE_iU^{-1})$, so $h_i \geq 0$, $h_i^2 = ah_i$, $\|h_i\| = a$ and $h_i|\Omega\rangle = 0$. Thus the parent is frustration-free and three-site local; overlapping terms need not commute. For fixed neighbours the local null vector is proportional to $(e^{bS/2}, e^{-bS/2})$. At $b = 0$, $h_i = a(I - X_i)/2$ and the ground state is the product of plus-X vectors.

Unitary equivalence gives the exact gap of A_N, with nonzero eigenvector UM. A supplied action $K > 0$ defines $H_N = KA_N$, energy gap

$$\Delta_N = Ka[1 - \tanh(2b)], \qquad e^{-tH_N/K} = Ue^{tQ_N}U^{-1}.$$

This is an imaginary-time identity in the chosen clock. The mathematical unitary $e^{-itH_N/K}$ supplies a different real-time law, not the original stochastic transition law. Neither the square-root amplitudes' physical state interpretation nor a physical measurement rule follows from the weighted-space coordinate change. The classical Ising energy $-J\sum\sigma_i\sigma_{i+1}$ is also a different operator from this parent.

Volume, coupling, global mixing and physical limits

Uniform bounds $K_N \geq K_0 > 0$, $a_N \geq a_0 > 0$ and $b_N \leq B < \infty$ give $\Delta_N \geq K_0a_0[1 - \tanh(2B)]$ for the finite-volume sequence. Finite range alone is insufficient. At $b_N = (\log N)/4$ and fixed a, the relaxation gap is $2a/(N + 1)$. A fixed total refresh budget replaces Q_N by Q_N/N and gives $a[1 - \tanh(2b)]/N$ even at fixed coupling. Vanishing K_N separately closes the energy gap. These are different parameter limits, not a fixed-coupling thermodynamic failure.

At fixed finite a, b and fixed global total-variation tolerance, Lubetzky–Sly's Corollary 3 gives

$$t_{\mathrm{mix}}^{(N)}(\epsilon) \sim \frac{\log N}{2a[1-\tanh(2b)]}.$$

The exact gap match (preprint p. 4) and mixing-time statement were read in B73, using the primary preprint with the rate-a conversion stated here. A uniform slow-mode rate can coexist with logarithmic global equilibration time. Neither constructs an infinite-volume Hilbert representation or a continuum quantum field.

The ground vector and locality survive $Q \mapsto \lambda Q$ while every rate and parent gap scales by λ. Independently changing K changes energy normalization. Thus equilibrium weights, a physical time law and a positive universal action unit remain separate inputs. The finite-depth cluster Hamiltonian is a distinct unitary-conjugation construction, not a second derivation of this stochastic parent. The field comparison states the further physical-operator and continuum obligations.

Consequence for STATE

The interacting benchmark supplies an exact all-mode finite-volume gap and an explicit local parent with known clock dependence. Its source evidence and full proofs are maintained together. No further Ising spectrum is required to understand the unresolved identification of a sampling generator, physical Hamiltonian, time unit and physical $\hbar$.